Memory location	Program	Comments
1008	MOV AL, 0FFH	Load 00H in accumulator
100A	OUT 0C8H, AL	Send through output port
100D	CALL DELAX (1100)	Call program to 1100.

Memory location	Program	Comments.
1100	MOV CX, 0505	Load 050H in Accumulator
1103	DEC CX	Decrement CX.
1105	JNZ 1104	Repeat ~~until~~ until Zero
1108	RET	Return to main Program.

Program : Sawtooth wave.

Assembly Program.

Memory location	Program Instruction	Comments.
1000	START: MOV AL, 00H	load 00H in accumulator.
1003	loop: OUT 08H, AL	Send through output port
1005	INC AL	Increment Contents of accumulator.
1007	JNC loop	Jump if no Carry.
1009	JMP START	go to starting location.

Result:

Thus the Sawtooth and Square wave form has been generated

Ex'4
12/2/26

Interfacting DAC.

Aim:

To write an assembly language Program in 8086 to generate Sawtooth and Square waveform using DAC.

Apparatus Required.

S.No	Item	Specification	Quantity
1.	Micro processor kit	8086	1
2.	Power Supply	+5 V DC, +12 V DC	1
3.	DAC interface Board.	-	1

Program: Square wave.

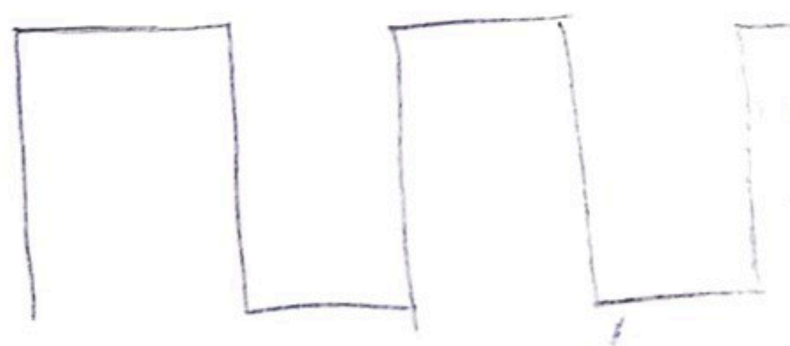
Memory location	Program	Comments
1000	MOV AL, 00H	load 00H in Accumulator.
1003	OUT 0C8H, AL	Send through output port.
1005	CALL DELAY (1100)	Call program to 1100.

TABULATION

waveform	Amplitude	Time period
Sawtooth	5V	1.48ms
Square	5V	9.8mS

MODEL GRAPH

Square wave:



Sawtooth wave: